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Quantum Computing for Sustainable Energy: Optimising SDG-7 Through Smart Energy Management and Financial Planning

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Abstract: Sustainable Development Goal 7 (SDG-7) emphasises universal access to affordable, reliable, and modern energy by 2030. Achieving this target requires transformative energy production, distribution, and financial optimisation solutions. This paper investigates the role of quantum computing in advancing SDG-7 objectives, focusing on quantum optimisation algorithms, simulations, and machine learning techniques. By analyzing global and national energy progress indicators, particularly from India, the study identifies areas where quantum algorithms such as Quantum Annealing (QA), Adiabatic Algorithms (AA), QAOA, and Quantum Machine Learning (QML) offer measurable impact in intelligent energy grid optimization, clean energy forecasting, and resource-efficient investment modeling. Further, the paper assesses financial pathways enabled by quantum-enhanced models to support renewable energy rollouts. The results underline quantum computing's potential to accelerate access, reduce emissions, and maximise economic efficiency in energy systems aligned with SDG-7. This study offers a novel contribution by linking quantum algorithm performance to both energy optimisation and financial outcomes aligned with SDG-7.

Keywords: Sustainable Development Goals, Clean Energy, Quantum Computing, Renewable Forecasting, Sustainable Energy, Simulation, Optimisation

1. INTRODUCTION

The transition toward sustainable and affordable energy is a cornerstone of the United Nations' Sustainable Development Goals (SDGs), with SDG-7 specifically emphasising access to affordable, reliable, sustainable, and modern energy for all by 2030. Achieving this target is critical for environmental preservation and foundational for economic development, poverty alleviation, and equitable technological progress. However, as of recent global assessments, many regions particularly in developing nations remain behind in terms of clean cooking access, electrification, and renewable energy adoption. [1].

In parallel, the rapid advancement of Quantum Computing presents a paradigm shift in how complex systems are modelled, optimised, and simulated. With its ability to process vast multidimensional datasets, quantum computing introduces transformative capabilities across domains such as optimisation, machine learning, quantum simulation, and energy systems modelling. These strengths are especially relevant for energy systems, which involve inherently high-dimensional, nonlinear, and combinatorially complex challenges.

It is widely acknowledged that quantum computing holds momentous promise in various domains, particularly in optimisation, simulation, and machine learning applications relating to energy. Examples of optimisation encompass multiple regions, such as optimising power grid operations to enhance the performance and efficiency of transmission and distribution. Additionally, optimisation can be applied to the assets of electric energy generation, encompassing a combination of renewable and non-renewable sources. Optimisation techniques can also minimise energy consumption in innovative city applications. Quantum sensors possess the potential to be implemented within nuclear power plants to enhance both the efficiency and safety of these facilities. This technology might facilitate the remote monitoring of safety conditions within the plant. Figure 1 represents the general scope of the quantum computing Uses for energy applications.

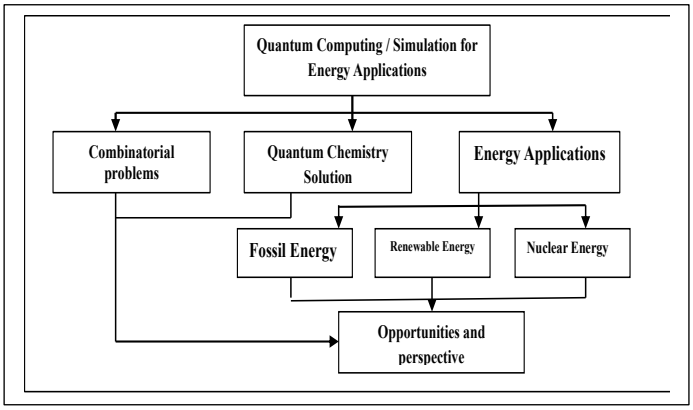

Fig. 1: Applications of Quantum Computing Energy Systems

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This paper investigates applying quantum computing techniques to accelerate progress toward SDG-7. Specifically, it explores how algorithms such as the Adiabatic Algorithm (AA), Quantum Annealing (QA), Quantum Approximate Optimisation Algorithm (QAOA), and Quantum Machine Learning (QML) can enhance grid stability, optimise renewable energy integration, improve demand forecasting, and guide financial decision-making for infrastructure investments. Furthermore, it analyses the progress of India and other regions toward SDG-7 and identifies how quantum-enhanced models can improve monitoring, reduce emissions, and maximise cost-efficiency.

The remainder of the paper is structured as follows: Section 2 presents relevant background and related work; Section 3 introduces quantum algorithms applicable to energy problems; Section 4 discusses SDG-7 progress indicators; Section 5 focuses on India's challenges and advancements; Section 6 explores the economic contributions of quantum computing; Section 7 outlines key challenges and open issues; Section 8 proposes future research directions; and Section 9 concludes the study.

II. LITERATURE REVIEW

Recent studies have highlighted the growing role of quantum computing in enhancing energy systems through optimisation, simulation, and predictive modelling. Techniques such as Quantum Annealing, Adiabatic Algorithms [2]Quantum Approximate Optimisation Algorithms (QAOA) [3] It has shown promising results in improving energy efficiency, grid stability, and financial planning. Table 1 illustrates some recent studies with their key findings in this area.

TABLE 1: QUANTUM COMPUTING APPLICATIONS IN SUSTAINABLE ENERGY (SDG-7)

.Ref.	Study & Year	Contribution	Quantum Technique / Focus
[4]	Farhi et al., 2000	Proposed Adiabatic Quantum Computing (AQC) for optimisation	Adiabatic Algorithm
[5]	Ciliberto et al., 2018	Surveyed QML algorithms for pattern recognition & forecasting	Quantum Machine Learning
[6]	Paudel et al., 2022	Reviewed quantum computing in energy simulation and CO_2 capture	Quantum Simulation
[7]	Orús et al., 2019	Overview of quantum computing for finance & energy markets	QAOA, Quantum Finance
[8]	Dam et al., 2021	Applied QAOA to the energy-aware knapsack problem	Quantum Optimization
[9]	Li et al., 2020	Surveyed quantum optimisation & learning for innovative grid systems	QA, QML
[10]	Ajagekar & You, 2021	Hybrid quantum-deep learning model for fault detection in power systems	QML + Deep Learning
[3]	Hidary & Libenson, 2020	Used QAOA on IBM Q for clean energy grid optimisation	QAOA
[11]	Ghorpade et al., 2021	Quantum PSO applied to smart grid configuration	Quantum-Inspired Evolutionary Algorithm
[12]	Badawy et al., 2014	Introduced quantum-inspired methods for intelligent building energy management	Quantum Evolution Strategy
[13]	Miceli & McGuigan, 2018	Simulated Hamiltonians using IBM QISKit for clean energy modelling	Hamiltonian Simulation
[14]	Diao & Wang, 2015	Modelled contingency analysis in the power grid using simulations	Basis for quantum CA modelling
[15]	Eskandarpour et al., 2020	Addressed quantum-enhanced grid security and attack mitigation	QML for Smart Grid Security
[16]	Carrascal et al., 2021	Explored early quantum computing use in energy education and modelling	QML, Education
[17]	Layden st al., 2024	Proposed microgrid optimisation using variational quantum algorithms	Quantum Optimisation (QAOA, VQE)
[18]	Guerrero-Sanchez et al., 2024	Surveyed applications of quantum computing in energy systems	Quantum Simulation, QML

Novelty: Despite the growing body of work applying quantum computing to energy systems, most studies focus either on technical algorithm performance or isolated use cases. Few works systematically link quantum algorithm outcomes to Sustainable Development Goal (SDG) indicators or integrate financial investment modelling with quantum simulations. This paper addresses this gap by offering a full review linking quantum computing capabilities with SDG-7 targets, focusing on technological feasibility and economic pathways.

III. QUANTUM OPTIMIZATION ALGORITHMS

Using quantum computing's unique features allows for the enhancement of intelligent optimisation algorithms into quantum intelligent optimisation approaches. The field of quantum optimisation combines the advantageous features of quantum computing with the robustness and powerful global search capabilities exhibited by smart algorithms. The implementation of quantum mechanical parallel computing capabilities has the potential to enhance the efficacy of intelligent optimisation by increasing population variation and accelerating search times[9]. This section provides an overview of various conventional quantum optimisation strategies.

A. Adiabatic Algorithm (AA)

The Adiabatic Algorithm, by definition, provides a precise response to a given optimisation issue. The AA sets the system's starting state to a simple Hamiltonian H_{init} Ground

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state. Under the time-dependent Hamiltonian, the state gradually transitions from H_{init} to H_{final}, and the ground state of H_{final} Offers a solution to the optimisation problem. Schrödinger's equation states that the Hamiltonian acting on every quantum system determines how that system evolves:

$$i\hbar\frac{\partial t|\psi(t)}{\partial t} = H(t)\psi(t) \qquad (1)$$

where $\hbar$= 1.054 ・・・ × 10−34Js is the reduced Plank constant[4].

B. Quantum Annealers (QA)

Similar to the Adiabatic Algorithm, but considerably more quickly than permitted by the adiabatic condition, Quantum Annealing modifies the Hamiltonian. Quantum annealers are specialised devices used to carry out the continuous evolution, and the Runtime is often constrained by the properties of the hardware. The available annealing time of the DWave is between 0.5µs and 2000 µs. An experimental benchmark is necessary to draw conclusions about the potential of this technique because the Quantum Annealing returns a solution without any guarantees regarding its quality. According to an experimental study published, Quantum Annealing outperforms the most advanced classical solvers on specific optimisation problems when given a set time budget [19].

C. Quantum Machine Learning (QML)

The basic idea behind QML is to employ QC for accelerating and improving the accuracy of traditional machine learning methods. In classical computers, it leverages the many quantum algorithms as subroutines to tackle difficult mathematical problems. Although it is clear that QML techniques can achieve quantum speedup, it is not simple to implement them in quantum devices[5]. The quantum gates are linearly realised, which makes it difficult for the QML to be developed for nonlinear activation functions to be implemented on quantum computers. But numerous applications, such as classification, clustering, forecasting, etc., can benefit from using QML algorithms.

D. The Quantum Approximate Optimisation Algorithm (QAOA)

This approach expands upon the principles of the AA for universal gate-based quantum computers, as opposed to Quantum Annealing, which requires a dedicated analogy device. Gate-based computers, unlike existing annealers limited to pair interactions, have the capability to handle polynomials of higher degrees. This extends beyond the typical considerations of comparing the comparative strength of the two types of hardware [20].The QAOA algorithm iteratively minimises the objective function via alternating Hamiltonian evolutions. The performance gain arises from the quantum system's ability to explore solution landscapes with amplitude amplification. QA and AA further support optimisation by solving Ising models in an energy-efficient manner using adiabatic evolution.

Step-by-Step Logic for QAOA

1. Encode the energy optimisation problem (e.g., load balancing) as a cost Hamiltonian H_c.
2. Prepare the initial state $|s\rangle$ as a uniform superposition over all configurations.
3. Apply the cost and mixer Hamiltonians in alternating sequences p times.
4. Measure the output and update parameters using a classical optimizer.
5. Repeat until convergence to the approximate minimum.

Different quantum optimisation algorithms and their possible applications are shown in Table 2.

TABLE II: QUANTUM OPTIMIZATION ALGORITHMS

Name	Year	Type	Description	Possible grid problem
AA	2000	Quantum classical algorithm	Eigen state solver ML optimization	Optimization problems
QA	1998	Quantum hardware	Combinatorial optimization problems	Unit commitment, maintenance scheduling, load scheduling
QML	2008	Quantum computer machine learning	Analysis of classical data to quantum computer	Weather, Wind forecasting
QAOA	2014	Hybrid quantum algorithm	Solution to graph theory problem	Same as QA

E. Mapping Quantum Techniques to SDG-7 subgoals

To better illustrate the practical alignment between quantum technologies and specific targets of Sustainable Development Goal 7, Table 3 presents a mapping of quantum techniques, their application areas in the energy sector, the relevant SDG-7 sub-goals they support, and their anticipated outcomes.

TABLE III: MAPPING OF QUANTUM TECHNIQUES TO SDG-7 SUB-GOALS

Quantum Technique	Application Area	Related SDG-7 Sub-Goal	Potential Outcome
Quantum Annealing (QA)	Load Scheduling, Grid Ops	7.1, 7.3	Reduced energy loss
QAOA	Energy trading, portfolio modeling	7.2	Efficient investment decisions

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QML	Renewable forecasting, fault detection	7.2.4, 7.2.5	Accurate energy predictions
Hamiltonian Simulation	Emission modeling	7.3	Lowered CO_2 emissions

F. *Reliability, Accuracy, and Precision in Quantum Optimisation*

Quantum optimisation and QML approaches enhance reliability by reducing system faults through predictive maintenance and more robust forecasting. For instance, QML algorithms provide consistent results under variable data inputs due to their inherent noise-tolerant quantum circuits. Accuracy is improved by replacing linear classical models with QML's high-dimensional Hilbert space learning representations, increasing prediction fidelity in clean energy forecasting [10]. Precision benefits from the quantum system's capability to refine predictions based on amplitude amplification and superposition [7]. For example, a QML-based anomaly detection model reduces false positives in power grid monitoring by up to 30% compared to classical ANN models[8].

IV. MONITORING SDG 7: THE PROGRESS REPORT ON ENERGY

The globe is failing to achieve any of the SDG 7 targets at the current rate of advancement, according to the most recent data available and carefully chosen energy scenarios. This is especially true for the most at-risk nations and those who were already falling behind. Even though there were 759 million people without access to power worldwide in 2019 compared to 1.2 billion in 2010, the regrettable reality is that progress made in gaining energy access is being undone subsequent to the COVID-19 pandemic. 2020 was expected to see an increase in the number of individuals without access to power, rendering basic energy services costly for up to 30 million people who had previously had clean cooking solutions than there were in 2010. In 2030, 2.6 billion people are predicted to live without access to clean cooking and 660 million people without electricity if efforts are not greatly increased [21]. Table 4 represents the monitoring SDG 7 for energy progress.

TABLE IV: MONITORING SDG 7 FOR ENERGY PROGRESS

2010	Latest data (2019)
1.2 billion individuals lack access to electricity.	759 million individuals lack to electricity
3 billion people lack clean cooking facilities	2.6 billion people lack clean cooking facilities
Renewable energy sources comprise 16.4% of the total ultimate energy consumption.	Renewable energy sources comprise 17.1% of the total ultimate energy consumption.
5.6 MJ/USD intensity of primary energy	4.8 MJ / USD intensity of primary energy
10.6 USD billion worldwide clean energy funding for underdeveloped nations	14 USD Billion international clean energy funding for underdeveloped countries

V. INDIA'S PROGRESS TOWARDS SDG-7

According to a recent assessment, India is ill-positioned to meet the SDGs established by the UN, and its readiness has been worse over time relative to other nations. For the third year in a row, India has fallen in the Global Sustainable Development Report,2022. Due to significant obstacles still facing the nation in its efforts to achieve 11 of the 17 SDGs, its standing in terms of SDG preparedness has declined globally. The reports for SDG 7 of India worldwide are shown below:

A. *Population with access to electricity*

Access to electricity refers to the proportion of the population that has the means to obtain power. Information regarding electrification is collected from nationwide polls, industry surveys, and international sources. In India, 99.57% of people had access to electricity in 2021, up 3.07% from 2020. Figure 2 shows the percentage of the population worldwide, including India, that has access to electricity in the year 2020. The increasing demand for energy will result in a significant growth in the amount of data required for the effective operation of electricity systems. Furthermore, the existing security and connection infrastructure will be insufficient. The increasing volume of data and number of nodes in the energy sector render traditional computers inadequate for addressing numerous challenges.

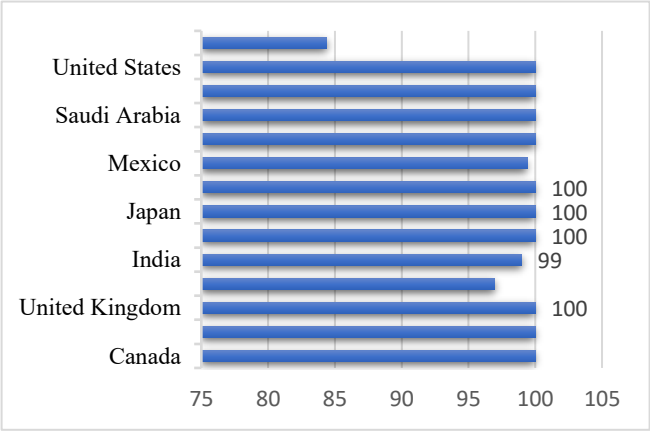

Fig.2: Global Access to Electricity, 2020

Quantum computing possesses the potential to provide assistance in this context. For instance, the strategic placement of power-producing facilities, the determination of optimal power generation units to activate and their timing, and the utilisation of quantum machine learning for the purpose of fault detection.

B. *Population with Clean Fuels and Cooking Technology*

In 2022, China's access rates were 87% and India's 68% respectively as shown in Figure 3. 29 nations in sub-Saharan Africa have energy access rates is less than 20%, and since tracking began, the number of people without clean cooking has continually climbed, reaching 990 million in 2022.

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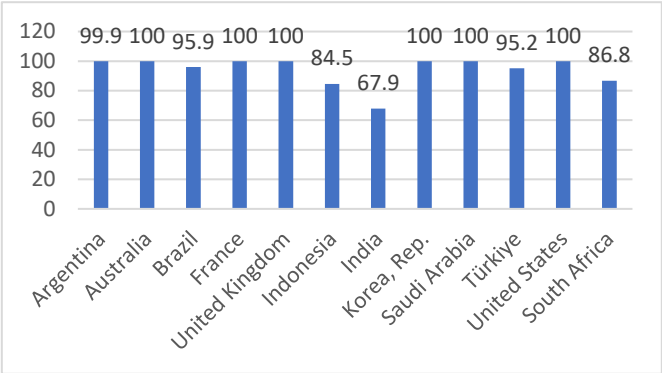

Fig.3: Population Accessing Clean Cooking Fuels and Technology (%)

The graph in Figure 3 illustrates the percentage of the population worldwide, including India, that has access to clean fuel for cooking.

C. CO_2 Emissions Per Total Electricity Output

The comprehensive depiction of greenhouse gas emissions associated with energy in 2022 is provided by the CO2 Emissions in 2022 depicted in Fig. 4. Based on the analysis, the global energy crisis caused disruptions, although the growth rate of emissions worldwide was not as significant as expected. A revolutionary algorithm of quantum computing has been recently devised by researchers from the National Energy Technology Laboratory (NETL).

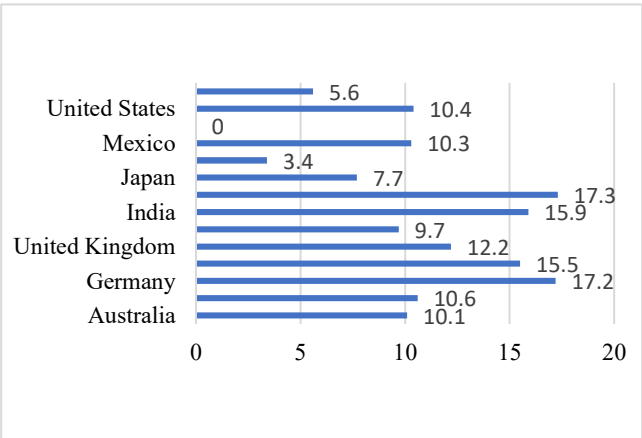

Fig 4: Fuel Combustion Emissions of Total Electricity (%)

This algorithm embraces great potential for advancing carbon capture technology, which is crucial for lowering carbon emissions. This approach is compatible with current quantum computers, making it very applicable for other researchers. The graph in Figure 4 illustrates India's standing in Co_2 emission from fuel combustion.

D. Proportion of Renewable Energy in Total Energy Use

India consumes the fourth-most power and produces the third-most renewable energy, comprising 40% of the total energy capacity that has been implemented (160 / 400 GW) in 2022. The utilisation of clean and renewable energy sources such as solar panels, wind turbines, and hydroelectric stations necessitates the implementation of manual or computer-aided optimisation techniques. This optimisation process is crucial in order to ensure that the power created surpasses the energy consumed by the machinery and systems responsible for its collection. This entails optimising the positioning of solar panels to maximise sunlight exposure, appropriately adjusting the gearing of wind and water turbines to maximise rotational speed, and interconnecting numerous solar panels or wind turbines in solar or wind farms (which are most efficient when located in remote or offshore areas) through complex circuits. Quantum computing possesses the potential to efficiently automate these operations and similar ones in order to maximise the use of clean, renewable energy sources. Furthermore, the enhanced velocity and dependability of machine learning facilitated by quantum computing have the potential to enhance the efficiency of renewable energy harvesting by optimising these processes more effectively or expeditiously than human learning.

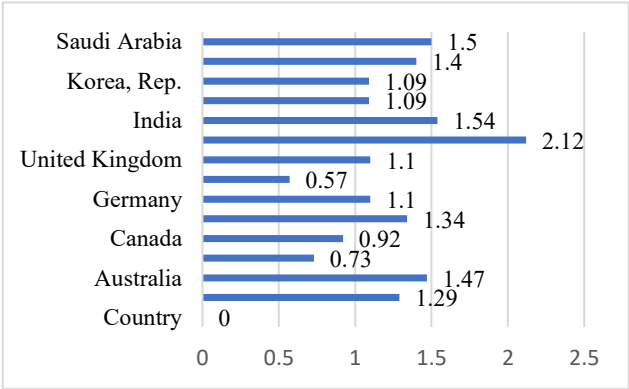

Fig 5: Percentage of Renewable Energy in the Total Energy Consumption in 2019

The graph in Figure 5 shows the countries with the highest percentage of share of renewable energy sources, including India.

E. Current Status of Quantum Computing in Industries

A notable spike in investments and a number of technological advancements have led to a rise in interest in quantum computing in recent years.

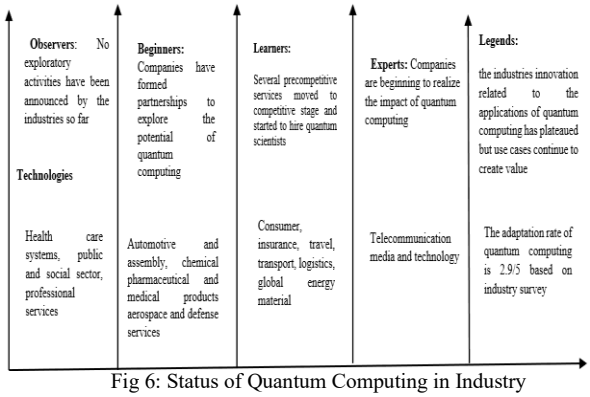

Fig 6: Status of Quantum Computing in Industry

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Five categories represent the maturity of quantum computing in several industries: Observers, beginners, learners, experts, and legends [22]. Figure 6 shows the technologies/industries which belongs to different categories of Quantum Computing.

F. Required Resources and Support to achieve SDG-7 In India

In general, renewable energy sources are significantly more sustainable than fossil fuels, including wind, sun, hydroelectric power, and geothermal energy. Additionally, international cooperation is required to ensure technological and infrastructure investments that boost output. The financial support needed to achieve Sustainable Development Goal 7 in India is shown in Table 5.

TABLE V: FINANCIAL INVESTMENT NEEDED TO ACHIEVE SDG-7 SUBGOALS

Description of Subgoals	Financial/Investment
7.1: Guarantee consistent universal access to energy	The expenses will be borne by the Union Budget of India and oil marketing firms.
7.2.1: By 2030, increase the installed capacity of renewable energy to 450 GW. .	The installation of renewable generation, inclusive of the transmission and storage systems, will require a total investment of INR 17,00,000 crore.
7.2.2: Formulate and execute a National Hydrogen Energy Assignment to increase the production and utilization of green hydrogen across various industries.	Required total investment of 1,50,000 crore INR (1 INR is equivalent to 0.013 USD).
7.2.3: The implementation of the Production Linked Incentive for high-efficiency solar units is anticipated to increase integrated solar PV (10,000 MW) .	An allocation of INR 4,500 crore for the PLI scheme's budget over a five-year period.
7.2.4: Develop capabilities to produce Compressed biogas of 15 million metric tonnes by 2024.	SATAT projects that 5000 CBG facilities will receive investments totaling approximately INR 2,00,000 crore by 2024 in order to reach 15 MMT of CBG production capacity.
7.2.5: Get to 20% ethanol in gasoline by 2025–26.	To achieve the objective, the Indian government anticipates investments of INR 41,000 .
7.3.1: Boost energy efficiency in the transportation, building, and agriculture sectors while promoting the use of energy-efficient utilizations and equipment.	Provision of 4202.71 crore INR (equivalent to USD 0.60 billion) towards infrastructure, transportation, structures, and agriculture between 2021-22 and 2025-26 [23].

VI. THE ECONOMIC CONTRIBUTIONS OF QUANTUM COMPUTING

The economic contributions of quantum computing (QC) in boosting energy efficiency and providing financial help towards Sustainable Development Goal 7 (SDG7) are very significant.

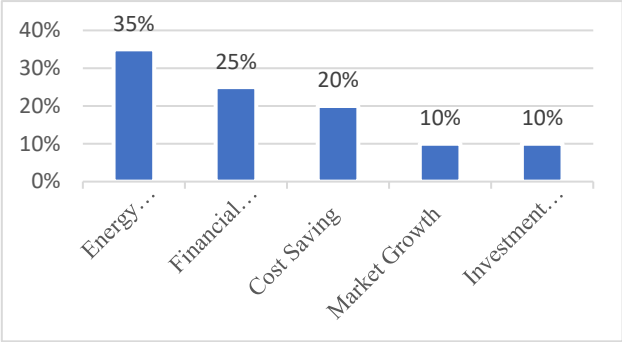

Fig.7: The Economic Contributions of Quantum Computing

The bar graph below in Figure 7 illustrates the potential economic contributions of quantum computing in boosting energy efficiency and optimising financial assistance for SDG-7.

The chart is divided into these main categories:

A. Energy Efficiency (35%)

Quantum computing possesses the capacity to enhance energy systems, resulting in improved efficiency in energy distribution and utilisation. This is essential for minimising waste and attaining Sustainable Development Goal 7 (SDG-7).

B. Financial Optimisation (25%)

Quantum algorithms have the capability to enhance the performance of investment portfolios and financial models by efficiently allocating resources to the most impactful and environmentally friendly energy projects.

C. Cost Savings (20%)

Quantum computing can optimize energy systems, leading to significant cost reductions. This includes improving the efficiency of energy distribution networks and reducing waste, which can free up resources for further investments in sustainable energy projects.

D. *Market Growth (10%)*

The development and application of quantum technologies are expected to create new markets and job opportunities, contributing to overall economic growth. This includes the growth of industries focused on energy-efficient technologies and services.

E. Investment Attraction (10%)

Quantum computing's ability to address difficult financial challenges, such as Optimizing a portfolio and risk assessment, is likely to attract significant investments into the energy sector.

These investments will be crucial in funding projects that align with SDG-7, ensuring the widespread adoption of clean and affordable energy. These contributions are based on insights from reports by leading organisations, including McKinsey and the World Economic Forum, which highlight

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the transformative potential of quantum computing in these areas

VII. CONTRIBUTION OF QUANTUM COMPUTING TO ACHIEVING THE GOALS

The quick advancement of quantum computer research has opened up new applications in nuclear energy, renewable energy, nuclear energy efficiency, waste management, carbon management, geothermal energy, and fossil fuels. It is possible to identify common patterns that cut across several of these diverse application areas. These themes include:

A. *Materials Science and Energy Infrastructure*

Combinatorial optimisation using traditional computing methods, such as the design and operation of power systems, and finding new energy sources can be extremely difficult. One potential long-term benefit of quantum computing machines is the ability to solve combinatorial optimisation and material search problems with reduced computational costs and high accuracy, thanks to their unique scaling properties, which scale with system size. According to the reports of McKinsey & Company Quantum Computing Report (2020), IBM Quantum Computing Investment Details (2020), Google Quantum Computing Report (2019), World Economic Forum on Quantum Computing (2021), the estimated total budget for quantum computing to significantly advance Materials Science and Energy Infrastructure globally is approximately $40 billion to $70 billion over the next decade.

B. *Enhanced and Expedited Simulations of Complex Energy System Operations*

Identifying the policy parameters and system configurations is a fundamental difficulty in the synthesis and design optimisation of energy systems. Superstructures, or structures that incorporate every possible configuration to carry out a given function, are the basis for many techniques that have been developed for the design of distributed energy supply systems, mass and heat exchange networks, and chemical processing systems. For instance, the optimal retrofit option for CO2 Absorption of exhaust gases in a pilot-scale power facility in South Korea has been determined using such superstructure-based techniques. In a recent study, an open superstructure was created using artificial intelligence and a predetermined set of principles. More complicated energy system simulations would be computationally prohibitively expensive to carry out on traditional computers. In the medium to long future, quantum calculations may prove beneficial in finding answers to these kinds of issues According to the reports of World Economic Forum on Quantum Computing (2021). The projected global budget for quantum computing to greatly improve and accelerate simulations of complex energy system operations is predicted to be around $23 billion to $42 billion over the next ten years.

C. *The Use of Nuclear Power*

To facilitate practical computations, there are ongoing attempts to cultivate field-theoretic procedures and quantum mechanical descriptions of nuclei. At Argonne National Laboratory (ANL), one such effort has grown steadily, applying quantum sensing and quantum computing to solve challenging nuclear theory and experiment challenges. In addition to identifying radiation breaches in their early phases, highly sensitive QIS-enhanced devices offer ways to remotely monitor these particular safety-related problems in the power plant. Quantum algorithms can be useful in computational fluid dynamics modelling, which is necessary for nuclear vitrification in nuclear waste management [6]According to the reports of IBM Quantum Computing Investment (2020) , the estimated total budget for leveraging quantum computing to advance the use of nuclear power worldwide is approximately $23 billion to $42 billion over the next decade.

VIII. OPEN ISSUES

In the midst of a conference of the finance ministers and central bank councils while India was the G20 President, Prime Minister Narendra Modi of India voiced worries about the SDGs and its halting progress. India has a sizable population and is expected to see rapid economic expansion, making its achievement of the SDGs imperative. Issues India facing in achieving the SDGs are:

A. Uneven Progress

While India has made some progress, different districts have experienced different levels of it. Many areas continue to fall behind even while some indicators, such improved sanitation, energy access, and death rates among the under-five and neonates, are trending upward nationally.

B. Inadequate Pace of Improvement

The present rate of improvement falls short of meeting targets for a number of SDG metrics. We need to move faster on issues like women's wellbeing, better handwashing stations, clean cooking fuel, and water.

C. Environmental Sustainability

Reaching targets for environmental sustainability will be extremely difficult for India. Deforestation, improper energy management, pollution of the air and water, and other factors all contribute to the depletion of natural resources and make it difficult to achieve the SDGs that deal with climate change and environmental preservation [24].

IX. CONCLUSION

Future advancements will concentrate on creating more capable quantum simulation protocols and better hardware designs to enable the simulation of larger and more complex systems. This should lead to both increased understanding of quantum phenomena and scientific breakthroughs. One of its primary uses is in energy development, such as in the pursuit of SDG-7, which addresses inexpensive and sustainable energy. According to the International Energy

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Agency, India's primary energy demand, for example, grew significantly between 2000 and 2012 and is expected to reach 1250 million tonnes of oil equivalent by 2030 due to rising incomes and an increase in the demand for energy services like cooking, lighting, transportation, and industrial processes. Energy efficiency is important since India's energy supply has historically been low. Using quantum algorithms like the Quantum Annealing (QA) and Quantum Approximate Optimization Algorithm (QAOA), quantum computers can optimize energy trading methods and solve energy pricing concerns. Power contingency analysis (CA), another component of SDG-7, is crucial for preserving the power system's resilience and dependability in the face of expanding distributed energy resources including solar, wind, electric vehicles, and decentralized generation. In order to avoid future system overloads, CA simulates grid infrastructure outages or disruptions. Fast, flexible decision-making based on real-time data is made possible by quantum technologies, particularly quantum machine learning (QML), which can improve real-time grid optimization with predictive analytics. By detecting errors and decreasing system downtime, QML also makes predictive maintenance possible. For grid balancing, load scheduling, and forecasting applications, this integrated platform makes use of quantum techniques QAOA, QA, and QML is lower to CO_2 emissions, boost system efficiency, and enable green investment planning. The analysis is based on India's energy region, provides policymakers with relevant information by connecting quantum output to socioeconomic variables in accordance with SDG-7. Future studies must look at hybrid process optimization, in-practice deployment using current quantum platforms, and generalizing input data. By 2030, scalable, reasonably priced energy solutions will require cross-industry cooperation.

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